

CSCE 640 QUANTUM ALGORITHMS · MODULE 04 · LECTURE NOTES

Interference

The one resource, isolated.

MODULE OUTCOMES

- Explain interference as the source of every advantage.
- Trace the amplitude cancellation in a concrete circuit.
- Explain why which-path information destroys it.
- Relate decoherence to the loss of interference.
- Use the cancellation requirement as a design principle.

1 The mechanism

A QUANTUM ALGORITHM ARRANGES MANY COMPUTATIONAL PATHS TO THE SAME OUTCOME, WITH PHASES CHOSEN SO THE WRONG ANSWERS CANCEL

Each basis state's final amplitude is a sum over every path the circuit provides to reach it — which is literally what the matrix multiplication computes, and is worth seeing as a path sum rather than as a vector update.

And the terms of that sum carry phases, so they can add constructively or destroy one another — which a sum of probabilities cannot do, because probabilities are non-negative (Module 02 §1).

So the entire design problem is: choose a circuit whose path phases make the unwanted outcomes' sums vanish, leaving the amplitude concentrated where you want to measure — and that is all any quantum algorithm does.

Which immediately explains the shortness of Module 01 §3's list — you need problem structure that lets you engineer the phases, and most problems do not have any: for an unstructured problem there is nothing to build the cancellation out of (Module 06 §4's lower bound is this intuition made rigorous).

$$H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}; \quad H|1\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$$

- The minus sign is the whole resource. H is the gate that turns a basis state into a pair of paths, and turns a pair of paths back into a basis state — which is why nearly every algorithm opens and closes with a layer of it.
- $H H |0\rangle = |0\rangle$: two paths reach $|1\rangle$ — via the $|0\rangle$ branch with sign $+$, and via the $|1\rangle$ branch with sign $-$ — and they cancel exactly, while the two paths to $|0\rangle$ reinforce.
- $H X H |0\rangle = -|1\rangle$: insert anything that flips the relative sign between the paths and the cancellation moves to the other outcome — which is the whole trick of the subject, at the smallest possible scale.

- **Trace this by hand; it is the smallest complete example of the mechanism**, and every algorithm in Modules 05 through 08 is this structure with more paths and cleverer phases.

2 Which-path information

IF ANYTHING RECORDS WHICH PATH WAS TAKEN, THE PATHS NO LONGER INTERFERE

Interference requires that the paths be indistinguishable at the point where they meet — if any leftover qubit, anywhere, is correlated with which path ran, the amplitudes add as probabilities instead and the cancellation does not happen.

Which is exactly Module 03 §2's garbage problem — an uncomputed ancilla is a which-path record — and leaving it in place is why a correctly derived algorithm silently stops working when implemented carelessly.

And it is also precisely what decoherence is — the environment records which path ran, accidentally and irreversibly — so noise and uncleaned garbage are the same failure arising from different causes, which is a genuinely unifying observation.

So the design rule is a single statement: nothing outside the computation may learn which path was taken, until the final measurement — which covers Module 03 §2 and the whole of Module 10 at once, and is worth carrying as the subject's one safety property.

3 Decoherence

- **Any coupling to the environment leaks which-path information — so the qubit must be isolated — and isolation conflicts directly with being able to control and read it**, which is where the difficulty lives.
- **Which is the central engineering tension of the whole field: controllable means coupled, and coupled means decohering — and every hardware platform (superconducting, trapped ion, photonic, neutral atom) is a particular position on that trade**, with different coherence times and different gate speeds.
- **So coherence time bounds circuit depth — the computation has to finish before the information leaks out —** which is why depth is one of Module 03 §4's five metrics and not an afterthought.
- **And it is not a matter of better shielding — the required isolation grows with the number of qubits and with the depth**, so a fixed engineering improvement buys a fixed factor against an exponential need — **which is why error correction rather than isolation is the answer** (Module 10).
- **Which makes Module 10's threshold theorem the result the entire field depends on:** it says that below a certain physical error rate, arbitrarily long computations become possible — that the problem is solvable in principle, which was not obvious and is not cheap. **Controllable means coupled, and coupled means decohering — which is why this is hard in a way better engineering alone does not fix.**

4 As a design principle

THE QUESTION TO ASK OF ANY QUANTUM ALGORITHM IS: WHAT CANCELS, AND WHY

For Deutsch-Jozsa, the constant case's paths all reinforce on the all-zeros outcome and the balanced case's cancel there (Module 05 §2) — **the cancellation does the deciding**, in a single query.

For Grover, each iteration rotates the state's amplitude toward the marked basis state by a fixed angle (Module 06 §2) — **and keeps rotating past it if you overshoot, which is the clearest available demonstration that this is geometry rather than accumulation** (and is what Project 1 asks you to plot).

For Shor, the Fourier transform makes the amplitudes at frequencies that are not multiples of the period cancel (Modules 07 and 08) — **which is the one case where the structure being exploited is entirely obvious** once stated: periodicity is exactly what a Fourier transform detects.

So if you cannot say what cancels, you have not understood the algorithm — and, more usefully, **if a proposed algorithm has no cancellation story, it has no advantage, which is a quick and effective filter on claims** (Module 13 §2).

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Preskill's notes, the lectures on quantum information (free)	§§2 and 3 — decoherence and which-path information, developed properly rather than by analogy.
Nielsen & Chuang, chapters 6 and 8 (library copy)	§1's path view and §3's open-system formalism.
Zurek — Decoherence and the transition from quantum to classical (free)	§§2 and 3 — the which-path account of decoherence from the person who developed it.
Aaronson — lecture notes on the source of quantum speedups (free)	§§1 and 4 — and the 'what cancels' question treated as the central one, which it is.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Write the amplitude of one outcome as a path sum** for a three-gate circuit.
- 2 **Trace $H H |0\rangle$** by hand, naming both paths to each outcome.
- 3 **Trace $H X H |0\rangle$** , and say which sign moved.
- 4 **Build a circuit where the cancellation is exact**, and verify it in your simulator.
- 5 **Then entangle an ancilla with the path**, and observe the interference disappear.
- 6 **Explain why that ancilla is a which-path record.**
- 7 **Relate that experiment to decoherence**, in four sentences.
- 8 **State the isolation-versus-control tension** and give two hardware platforms' positions on it.
- 9 **For each algorithm in Modules 05 to 08, say what cancels.**
- 10 **Find a claimed quantum algorithm** and ask what its cancellation story is.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 State the mechanism in one sentence.
- 2 Why can probabilities not do this?
- 3 Why does the mechanism explain the shortness of the advantage list?
- 4 Trace the two-qubit example and name the cancelling paths.
- 5 What does H do, structurally?
- 6 State the which-path rule.

- 7 Why is uncomputed garbage a which-path record?
- 8 How are garbage and decoherence the same failure?
- 9 State the isolation-control tension and its consequence for depth.
- 10 What is the question to ask of any quantum algorithm?